

Faculty Of Engineering and Technology

Electrical and Computer Engineering Department

ENEE 4113, Communication Laboratory

Amplitude Shift keying (ASK)

Pre-Lab Experiment #10

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Section: 5.

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Message Signal

$\boldsymbol{m(t)} = \boldsymbol{Am} * \boldsymbol{Square}(\boldsymbol{2\pi fmt})$ (Note: the parameters of the message signal will change now I set Am = 5, fm = 1khz, D = 50%)

Block Diagram:

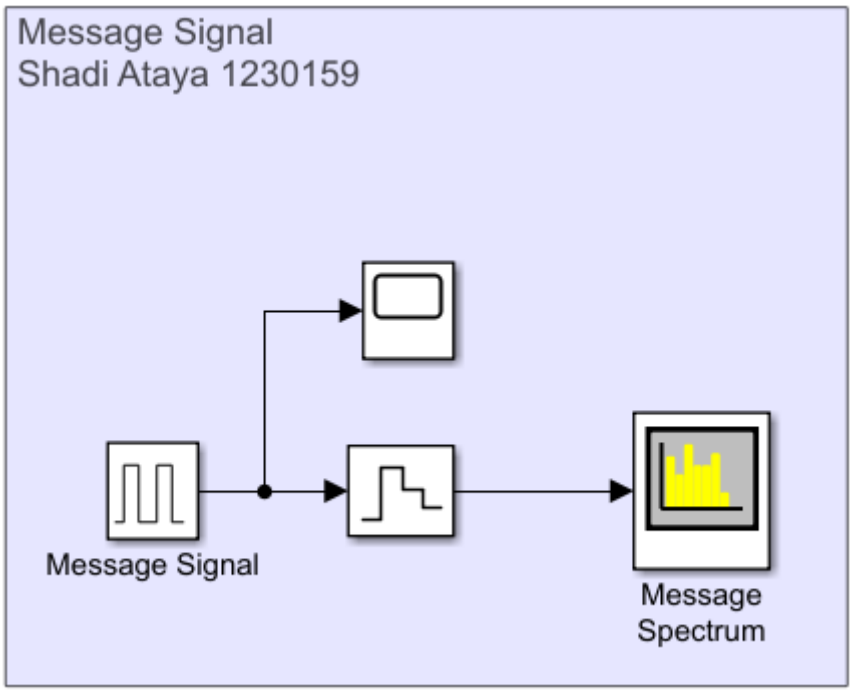


Figure 1: Message signal block diagram

Parameters:

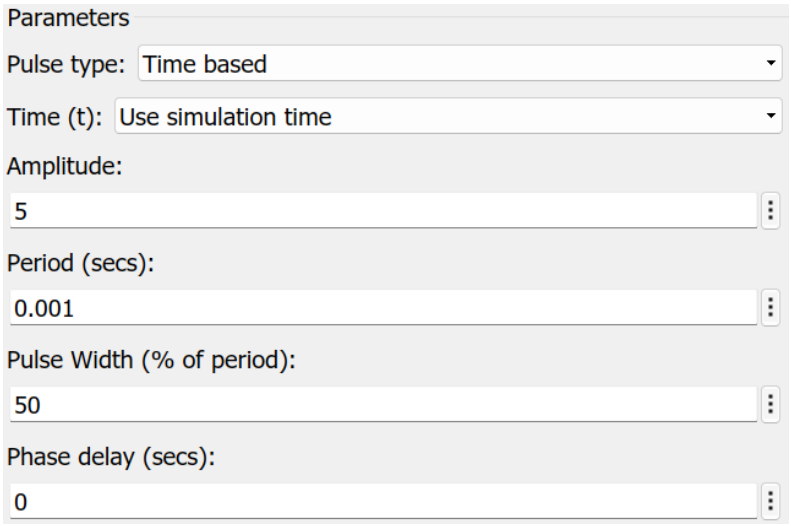


Figure 2: Message signal parameters

Message signal in time and frequency domain:

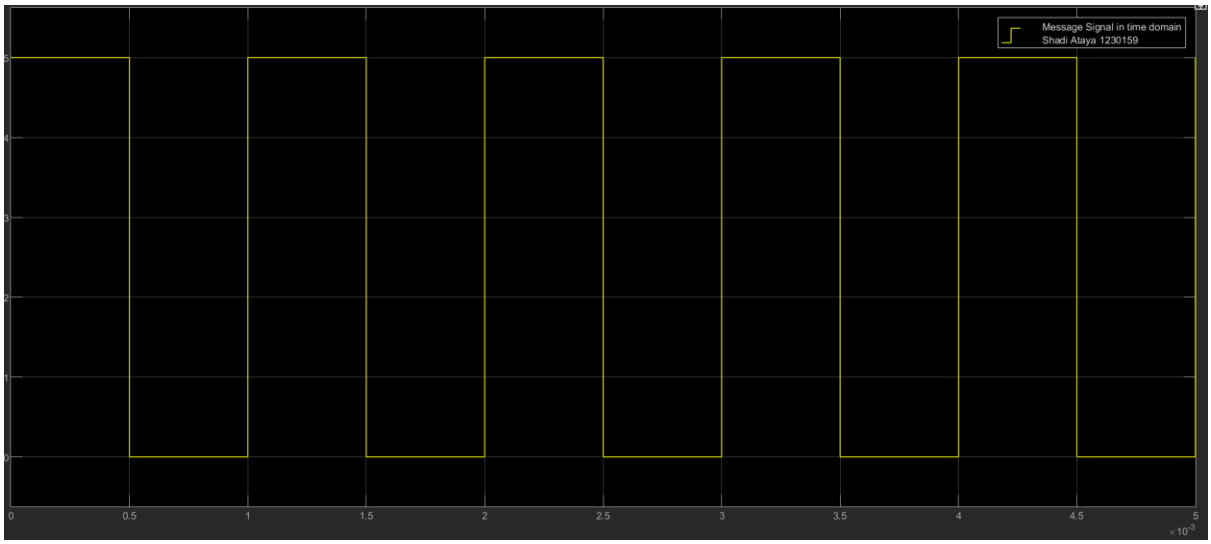


Figure 3: Message signal in time domain

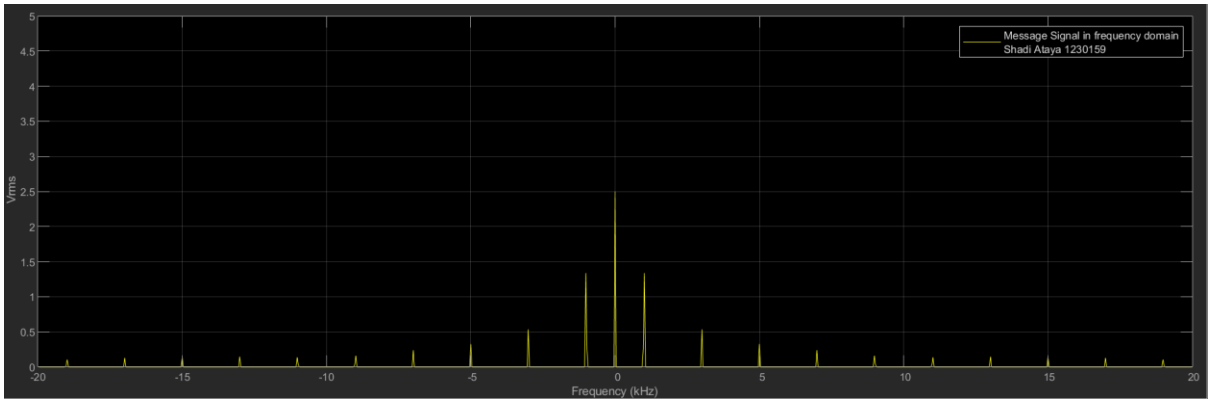


Figure 4: Message signal in frequency domain

Carrier Signal

$\mathbf{C(t) = cos(2\pi(20k)\ t)}$

Block Diagram:

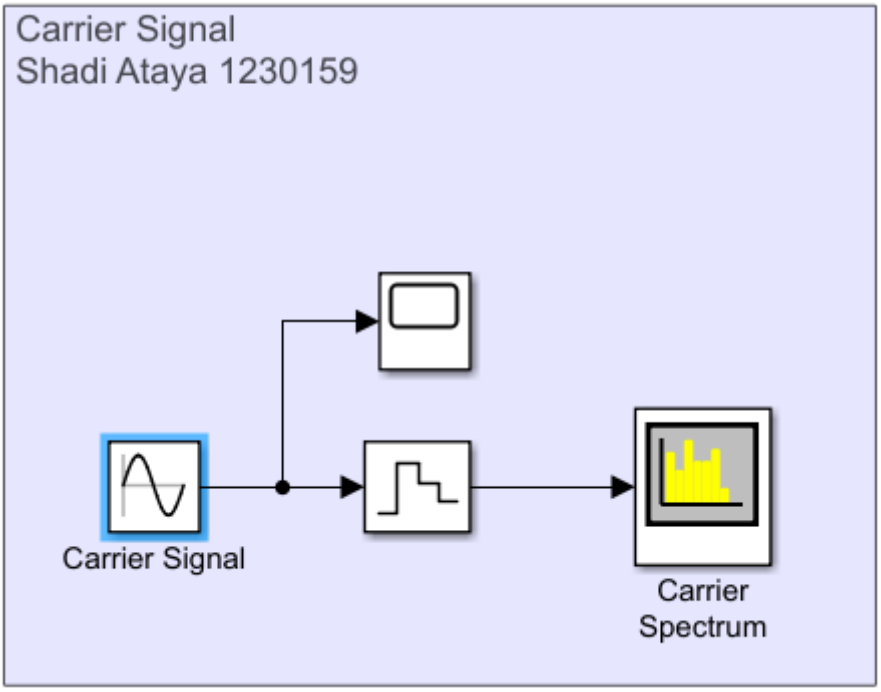


Figure 5: Carrier signal block diagram

Parameters:

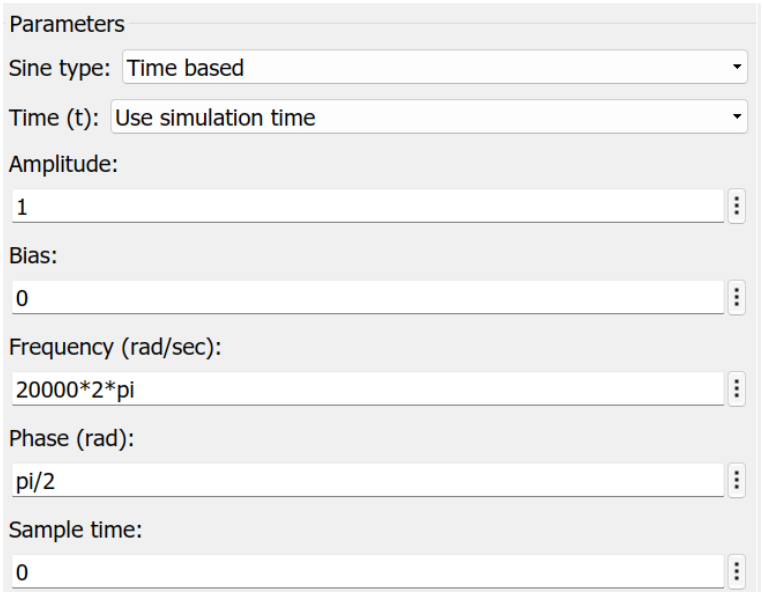


Figure 6: Carrier signal parameters

Carrier signal in time and frequency domain:

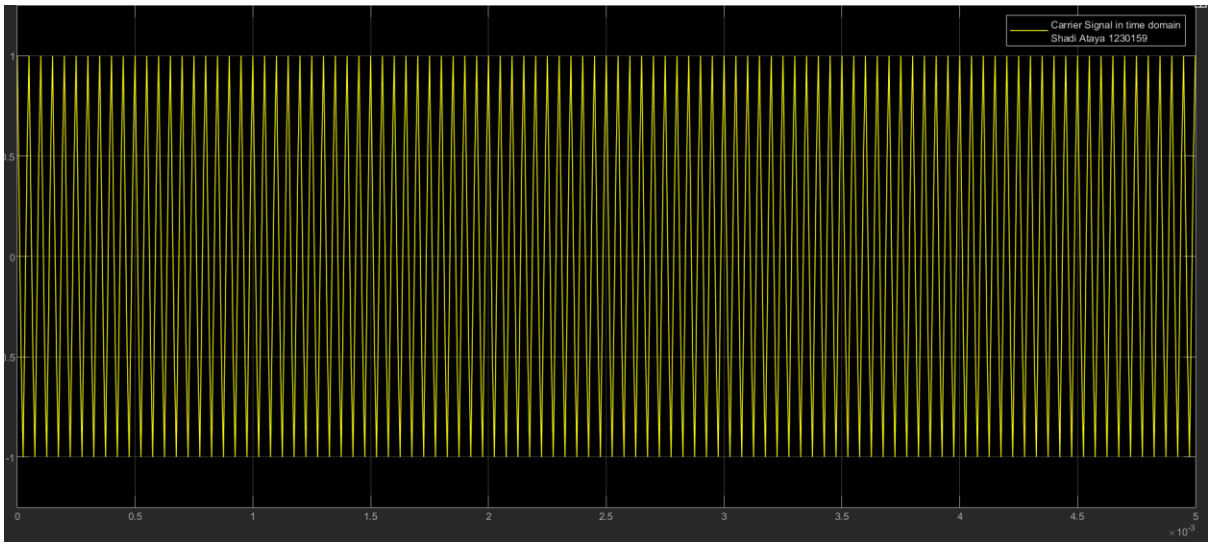


Figure 7: Carrier signal in time domain

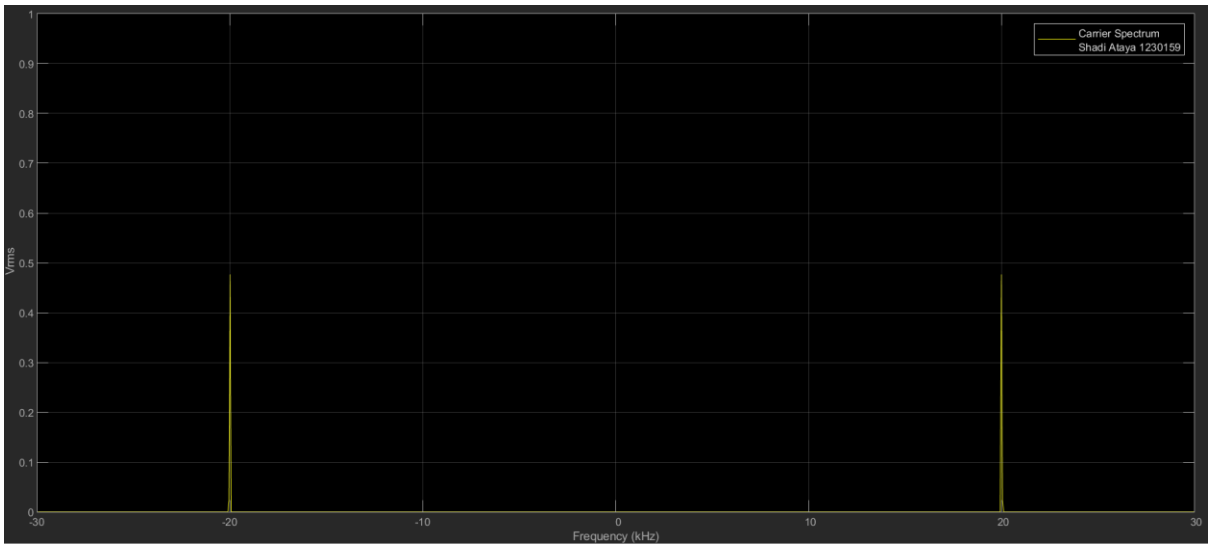


Figure 8: Carrier signal in frequency domain

ASK Modulation

Block Diagram:

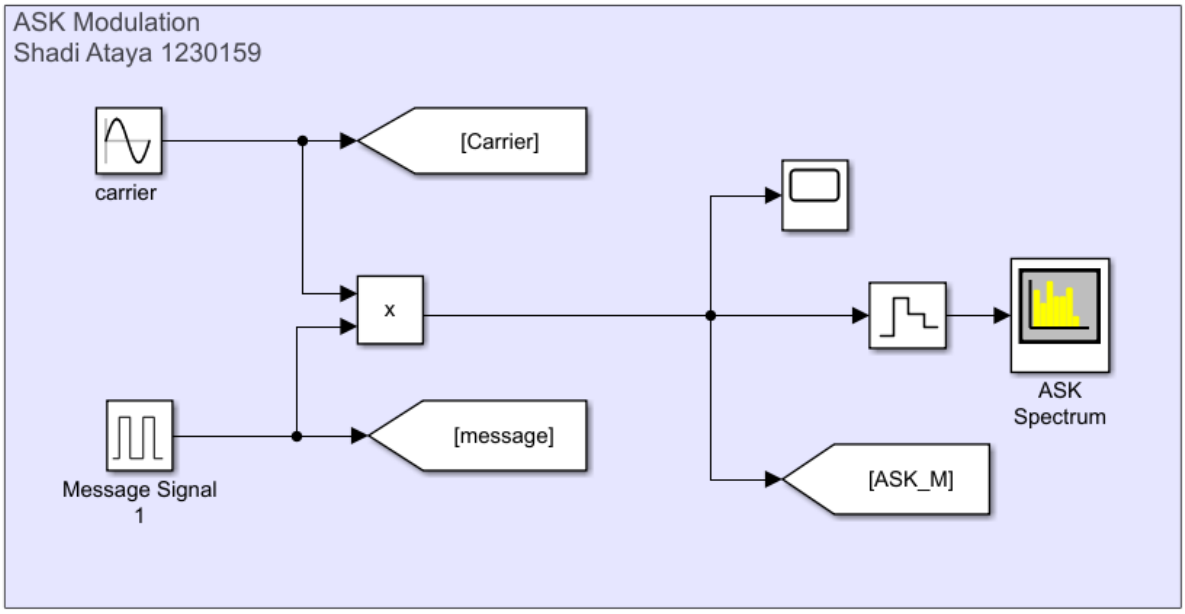


Figure 9: ASK Modulation Block Diagram

ASK Modulated signal @ Am=5, D=50% & f=1KHz

Time Domain:

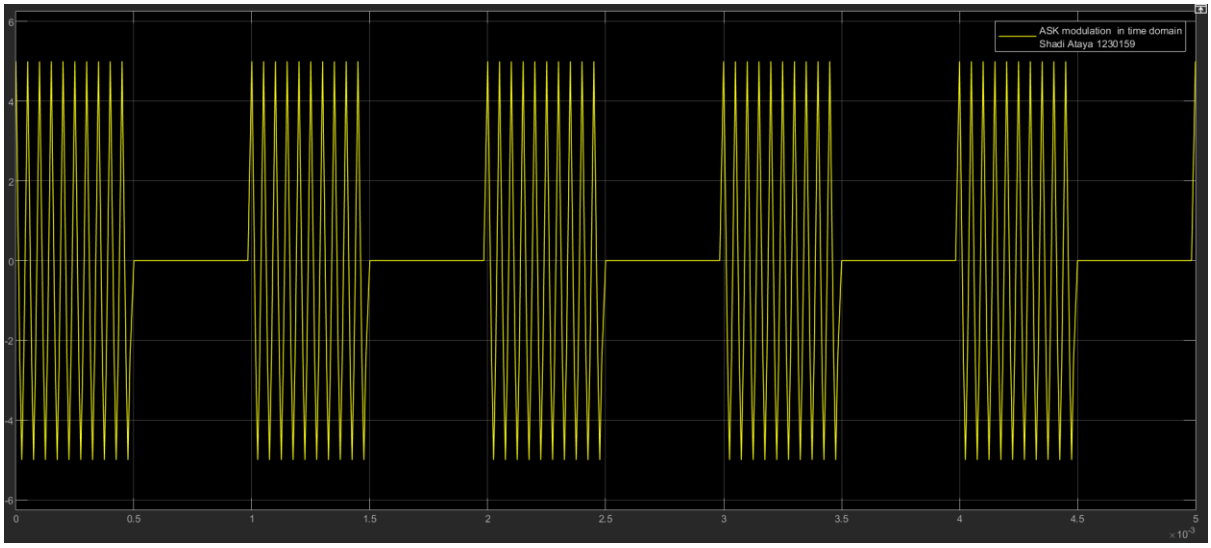


Figure 10: ASK Modulated @ Am=5, D=50% & f=1KHz in time domain

- The ASK output in the time domain consists of a 20 kHz carrier that turns ON and OFF according to the 1 kHz rectangular message pulse. Since the pulse has an amplitude of 5 and a 50% duty cycle, the carrier appears at ±5 during the ON interval (0.5 ms) and disappears to 0 during the OFF interval (0.5 ms). As a result, each ON window contains

about ten cycles of the 20 kHz carrier, followed by a flat zero region during the OFF window. This produces a distinct ON–OFF pattern where the amplitude of the carrier remains constant during the ON interval but completely vanishes during the OFF interval, and the overall envelope of the modulated signal perfectly follows the 1 kHz message pulse shape.

Frequency Domain:

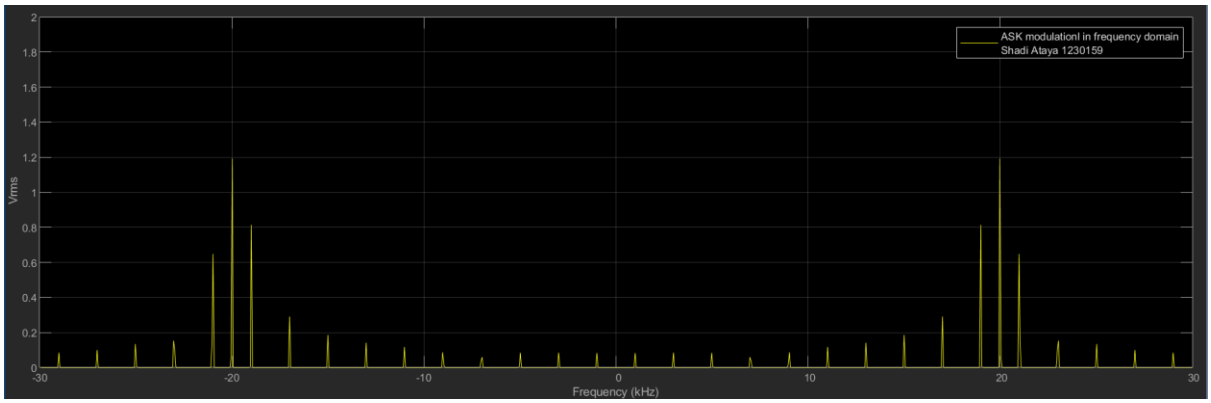


Figure 11: ASK Modulated @ Am=5, D=50% & f=1KHz in frequency domain

- In the frequency domain, the ASK signal shows a strong carrier component at 20 kHz, and the switching caused by the 1 kHz rectangular message introduces sidebands spaced at multiples of 1 kHz around the carrier. Because the message has a 50% duty cycle, its square-wave nature produces harmonic components that appear as additional sidebands around 20 kHz. As a result, the signal's main energy is concentrated at the carrier frequency, with multiple sidebands created by the periodic ON–OFF modulation of the carrier.

ASK Modulated signal @ DC signal 0V

To generate an ASK modulated signal with a pure DC level of 0 V in the ASK system, the message pulse was replaced with a constant block set to 0.

Block Diagram:

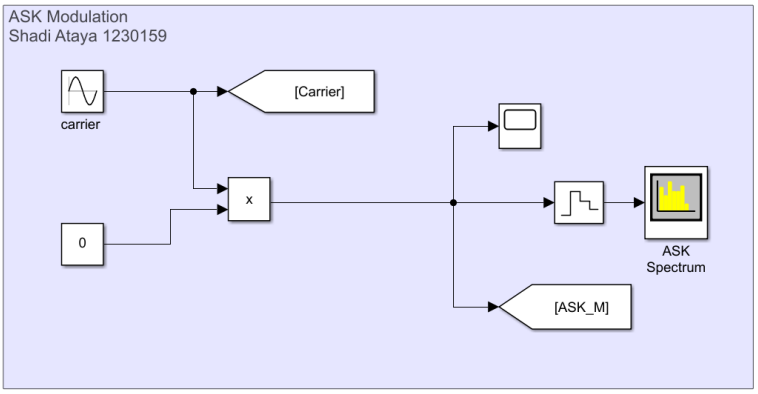


Figure 12: ASK Modulation Block Diagram with DC = 0 v

Time Domain:

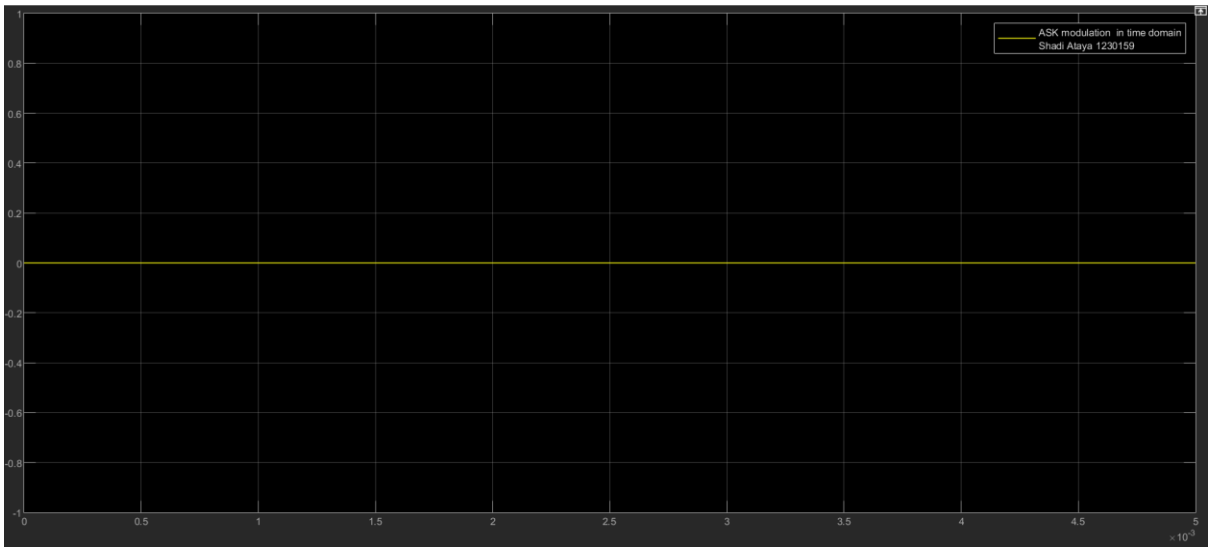


Figure 13: ASK Modulated signal @ DC signal 0V in time domain

- In the time domain, the ASK output becomes a flat line at 0 volts because using a DC level of 0 fully suppresses the carrier. With no amplitude applied to the 20 kHz carrier, the modulator produces no oscillations, resulting in a continuous zero-level signal that indicates the absence of transmission.

Frequency Domain

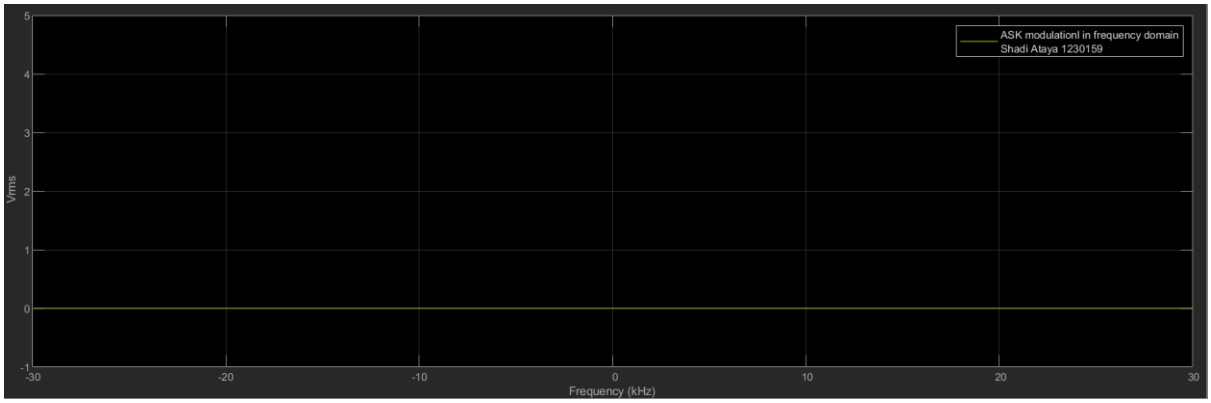


Figure 14: ASK Modulated signal @ DC signal 0V in frequency domain

- In the frequency domain, the signal contains no spectral components because the input is a DC level of 0 volts, which suppresses the carrier completely. With the output fixed at 0 V, no energy appears at any frequency, including the carrier, meaning the transmitter sends nothing at all.

ASK Modulated signal @ DC signal 2.5V

To generate an ASK-modulated signal with a DC level of 2.5 V, the message signal is replaced with a constant value of 2.5 V so that the carrier is scaled to a fixed amplitude of 2.5 throughout the entire duration.

Time Domain:

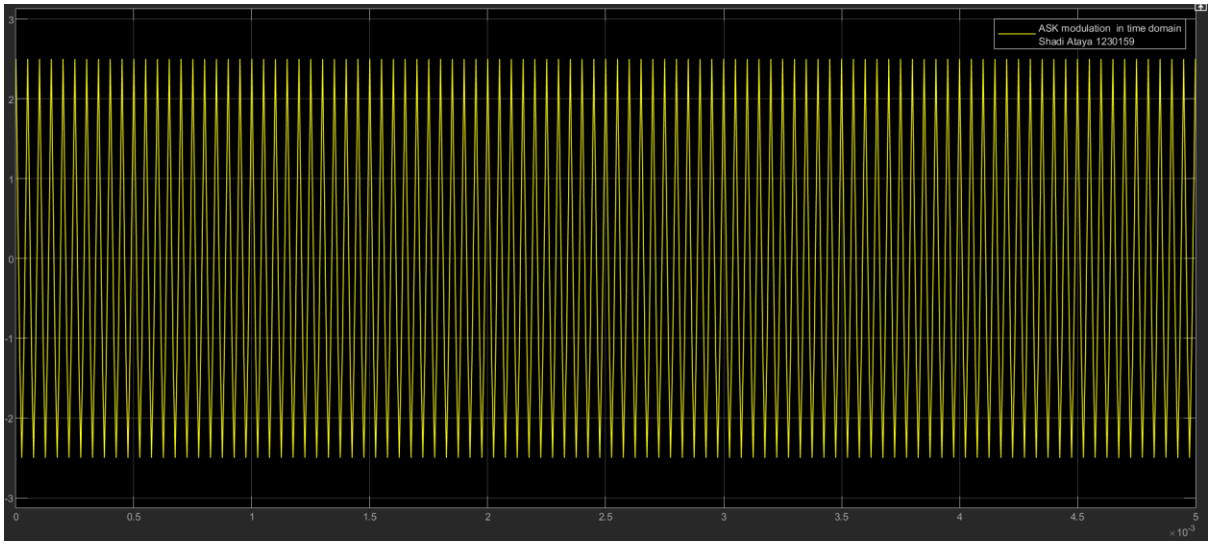


Figure 15: ASK Modulated signal @ DC signal 2.5V in time domain

- In the time domain, the ASK output becomes a continuous carrier wave with a fixed amplitude because the input is a DC level of 2.5 V. The carrier always maintains the same amplitude, with no ON–OFF changes or variations, so the signal does not encode any data and simply appears as a steady, unmodulated carrier.

Frequency Domain

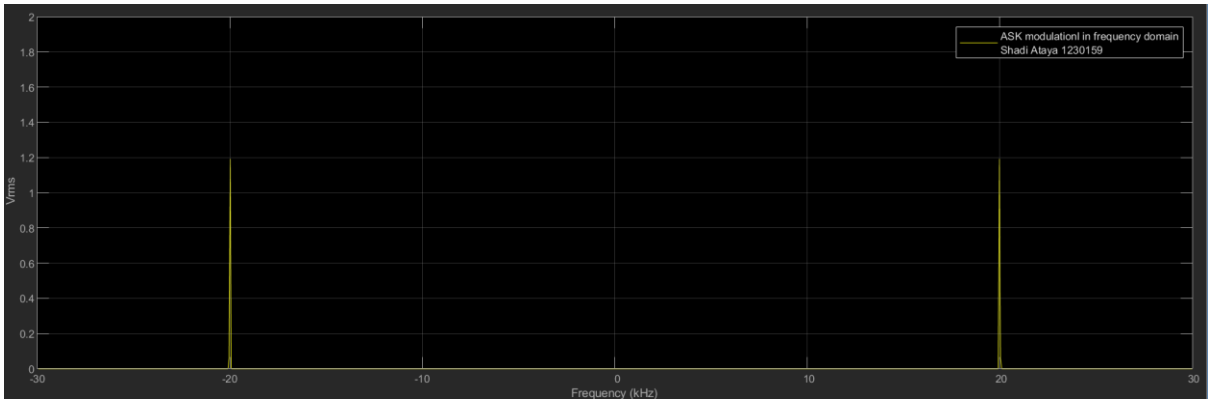


Figure 16: ASK Modulated signal @ DC signal 2.5V in frequency domain

ASK Modulated signal @ Am=1.5, D=50% & f=1KHz

Time Domain

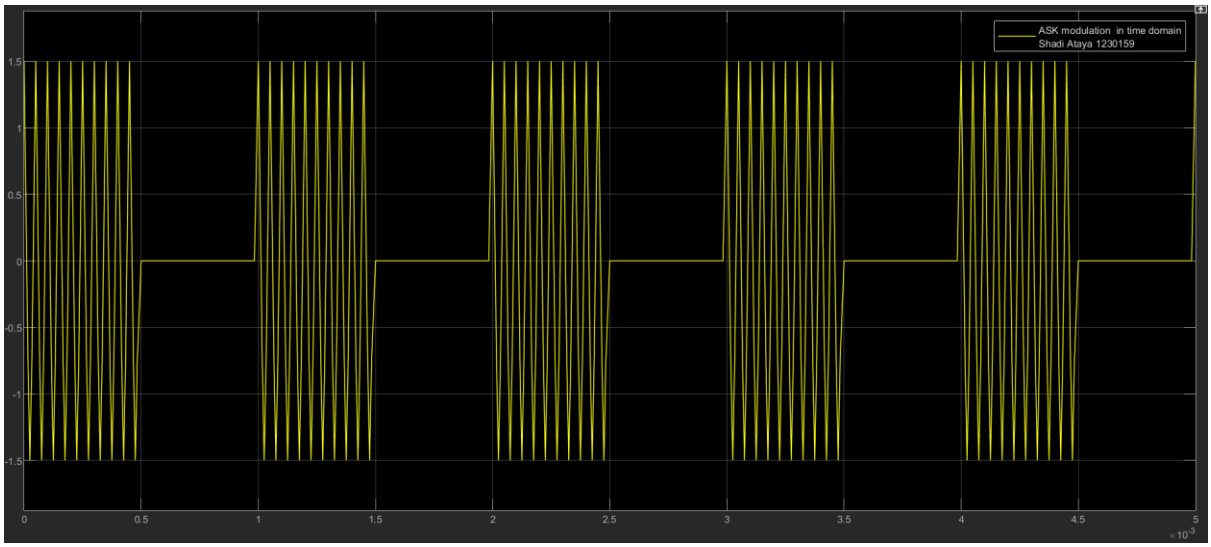


Figure 17: ASK Modulated signal @ Am=1.5, D=50% & f=1KHz in time domain

- In the time domain, the ASK signal keeps the same ON–OFF shape as (Am = 5, D = 50% fm = 1KHz), but the amplitude of the carrier during the ON intervals is reduced because the message amplitude is lowered from 5 V to 1.5 V. The carrier still appears only during the 0.5 ms ON window of each 1 kHz pulse, but its peak value is now ±1.5 instead of a ±5, while the OFF intervals remain at zero

Frequency Domain

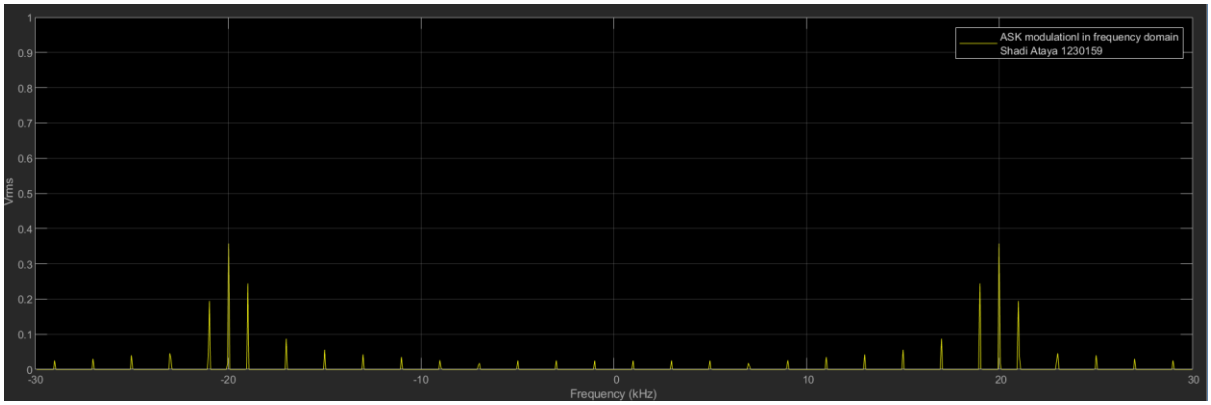


Figure 18: ASK Modulated signal @ Am=1.5, D=50% & f=1KHz in frequency domain

- In the frequency domain, the spectral components decrease in magnitude because the message amplitude is reduced from 5 V to 1.5 V, but the bandwidth remains the same since the modulation frequency and duty cycle are unchanged. The carrier and all sidebands retain their positions in frequency, only with lower amplitudes.

ASK Modulated signal @ Am=5, D=50% & f=0.5KHz

Time Domain

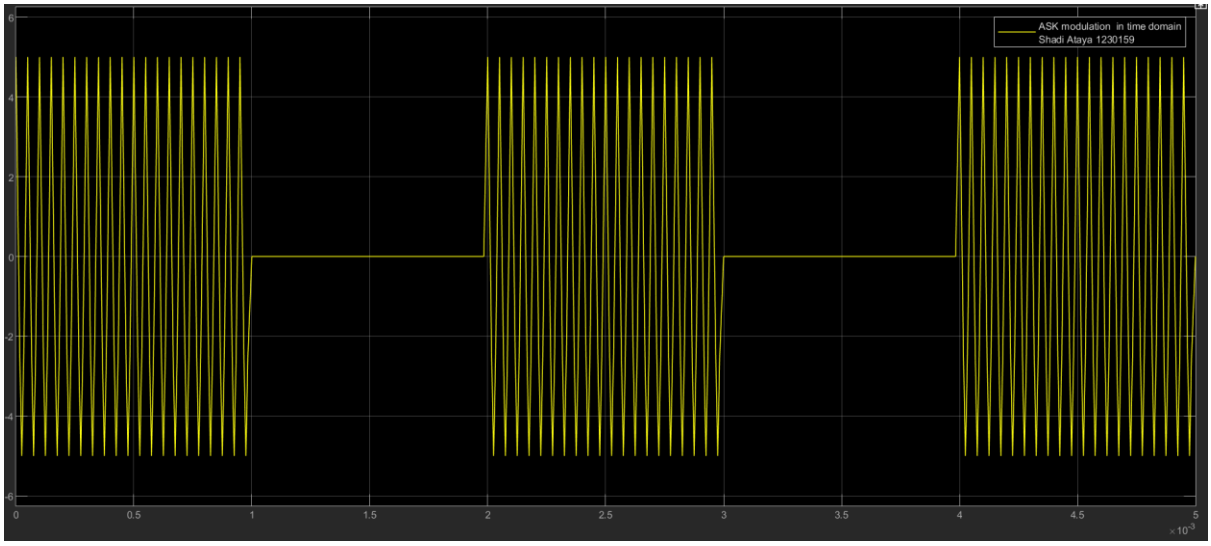


Figure 19: ASK Modulated signal @ Am=5, D=50% & f=0.5KHz in time domain

- In the time domain, lowering the message frequency to 0.5 kHz makes the ASK signal switch between its "high-amplitude" and "low-amplitude" states more slowly. Because the message period becomes longer, the modulated waveform shows clear and extended high-low-high-low amplitude sections. The carrier still oscillates rapidly at its original frequency, but the envelope changes more gradually, producing a very visible and distinguishable pattern. With an amplitude of 5 and a 50% duty cycle, each high and low section lasts the same duration, giving the ASK signal a clean and regular envelope that corresponds to the slower 0.5 kHz message.

Frequency Domain

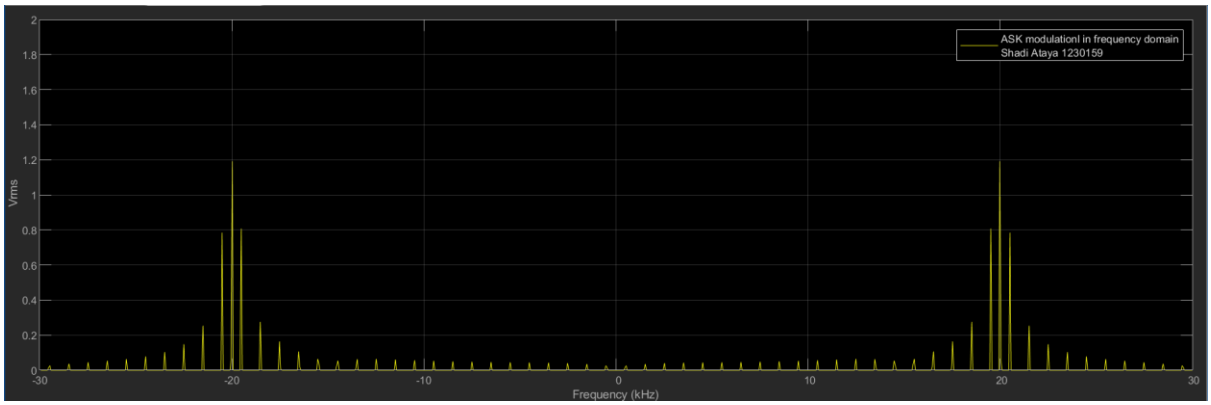


Figure 20: ASK Modulated signal @ Am=5, D=50% & f=0.5KHz in frequency domain

- In the frequency domain, lowering the message frequency to 0.5 kHz causes the sidebands to move closer to the carrier, so the main spectral components now appear at 20 kHz ± 0.5

kHz. The carrier at 20 kHz remains the dominant peak, while the sidebands become more closely spaced, reflecting the slower switching of the 0.5 kHz message. Because the sidebands are now tighter around the carrier, the overall bandwidth of the modulated signal decreases. The spectrum shows a clear carrier with symmetrical sidebands, demonstrating proper ASK modulation and making the structure of the signal easy to observe and interpret.

ASK Modulated signal @ Am=5, D=10% & f=1KHz

Time Domain

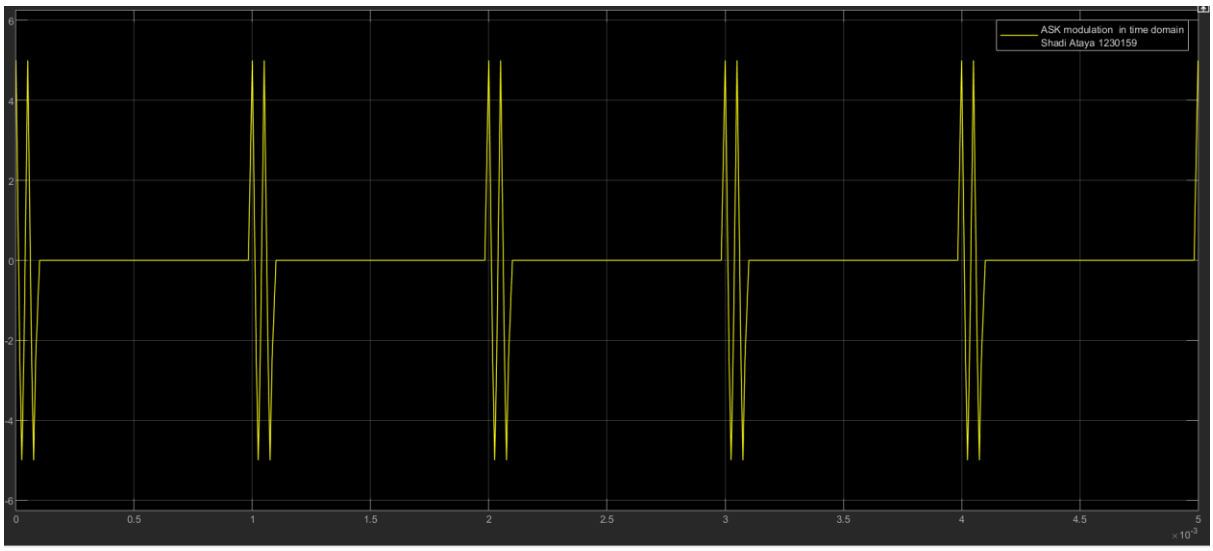


Figure 21: ASK Modulated signal @ Am=5, D=10% & f=1KHz in time domain

- In the time domain, reducing the duty cycle to 10% causes the ASK waveform to spend only a very short portion of each cycle at its high amplitude. As a result, the signal shows brief, narrow “high” bursts followed by long low-amplitude intervals. Although the carrier continues to oscillate at high frequency, its envelope becomes sharply pulsed, with short active sections and long inactive sections. The 1 kHz message frequency still determines the spacing between bursts, but the envelope now appears much more compressed because only 10% of each period carries the high amplitude of 5.

Frequency Domain

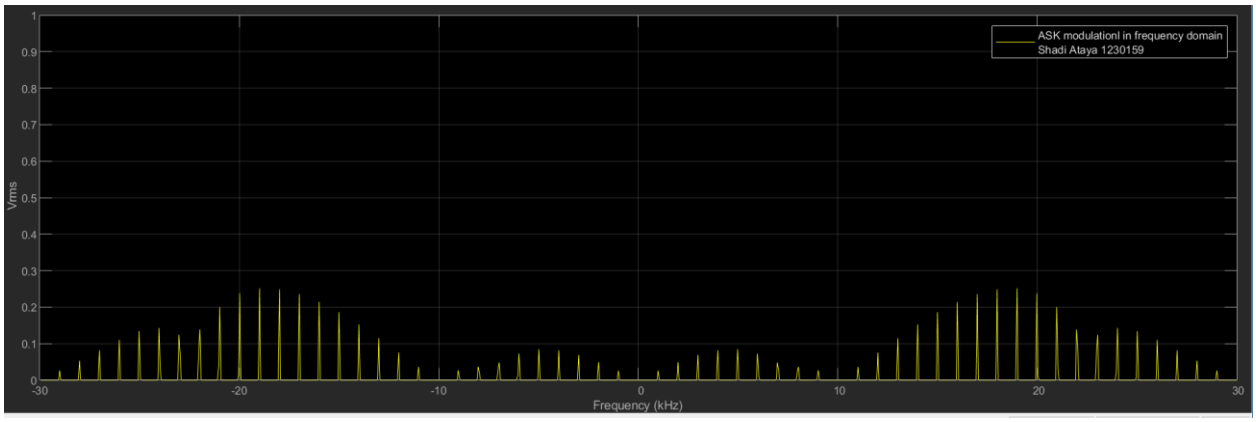


Figure 22: ASK Modulated signal @ Am=5, D=10% & f=1KHz in frequency domain

- In the frequency domain, reducing the duty cycle to 10% causes the ASK signal to exhibit a much wider spectral spread. Because the carrier is transmitted for only a very short portion of each cycle, the message envelope becomes a narrow pulse, and these sharp transitions introduce many high-frequency harmonics. As a result, the sidebands around the 20 kHz carrier extend farther outward and become more pronounced compared to the 50% duty-cycle case. The carrier at 20 kHz remains the strongest component, but the spectrum shows stronger harmonics, wider spreading, and therefore an overall increase in bandwidth, directly reflecting the abrupt on–off behaviour created by the 10% duty cycle.

ASK Demodulation 1

Block Diagram:

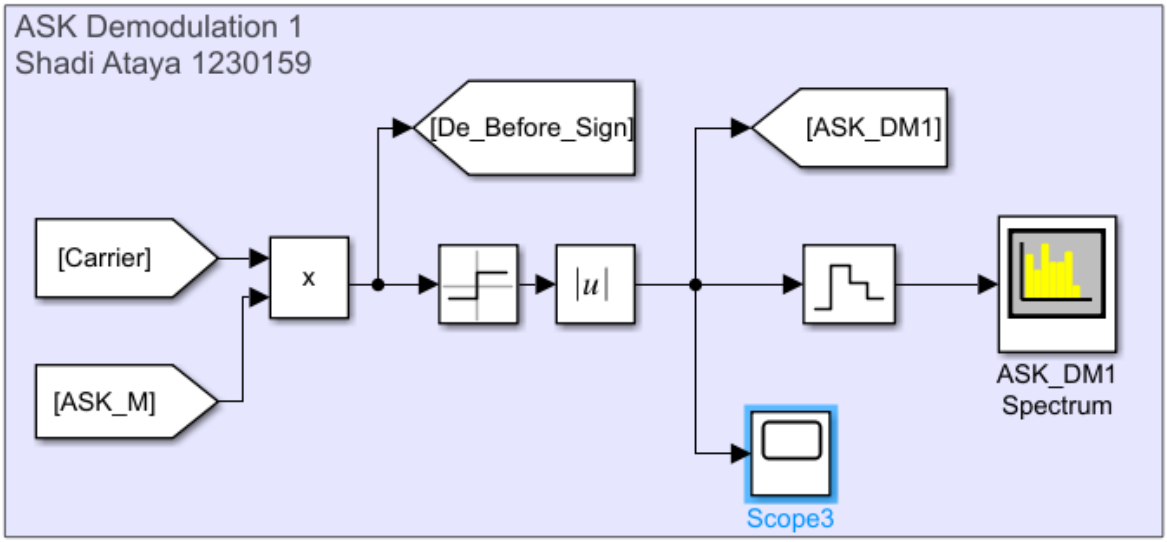


Figure 23: ASK Demodulation (1) Block Diagram

De-Modulated (1) signal @ Am=5, D=50% & f=1KHz

Time Domain

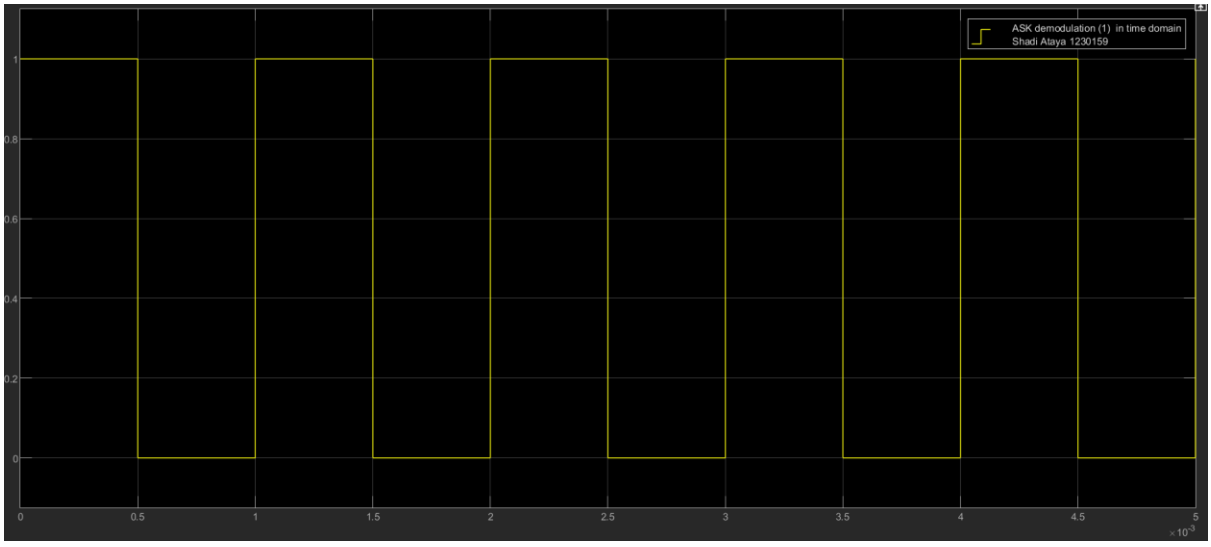


Figure 24: De-Modulated (1) signal @ Am=5, D=50% & f=1KHz in time domain

Frequency Domain

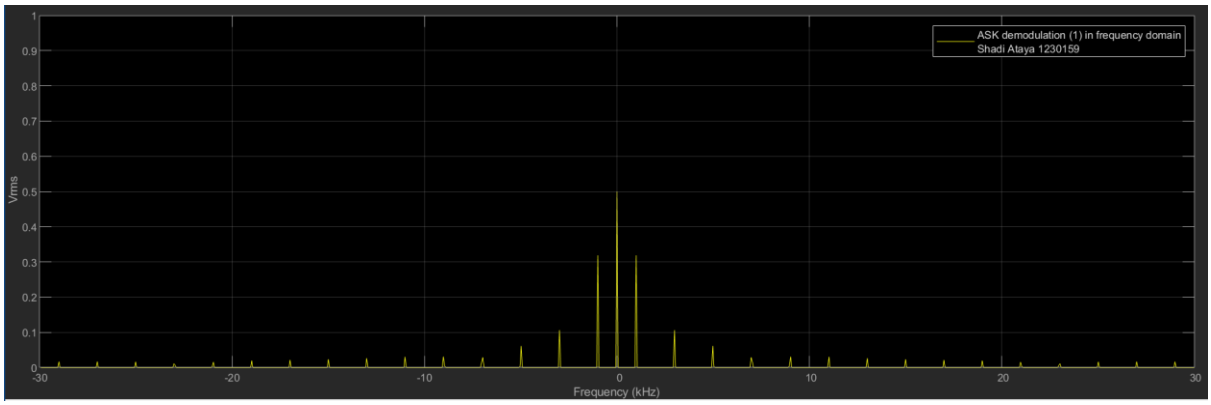


Figure 25: De-Modulated (1) signal @ Am=5, D=50% & f=1KHz in frequency domain

De-Modulated (1) signal @ Am=1.5, D=50% & f=1KHz

Time Domain

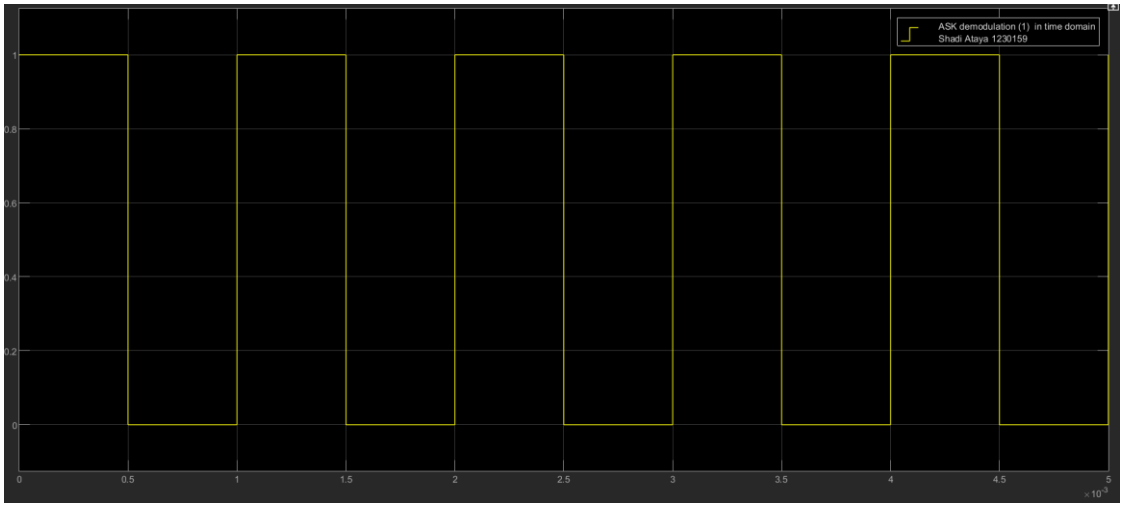


Figure 26: De-Modulated (1) signal @ Am=1.5, D=50% & f=1KHz in time domain

Frequency Domain

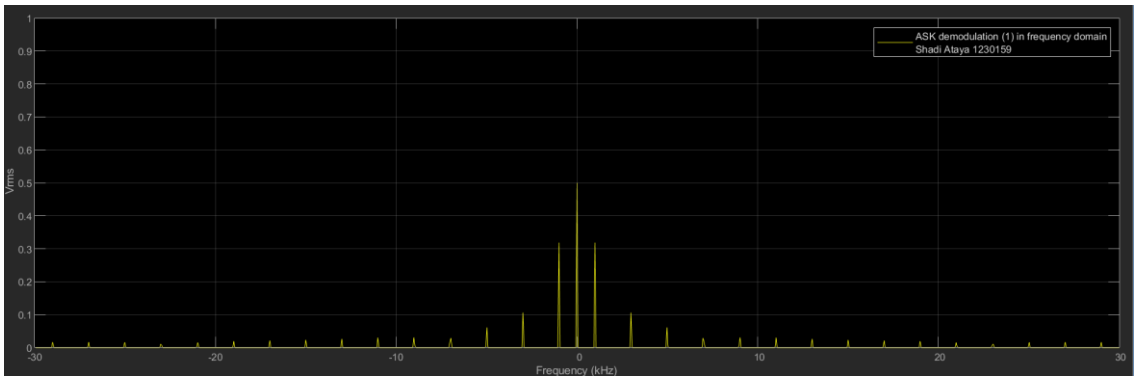


Figure 27: De-Modulated (1) signal @ Am=1.5, D=50% & f=1KHz in time domain

De-Modulated (1) signal @ Am=5, D=50% & f=0.5KHz

Time Domain

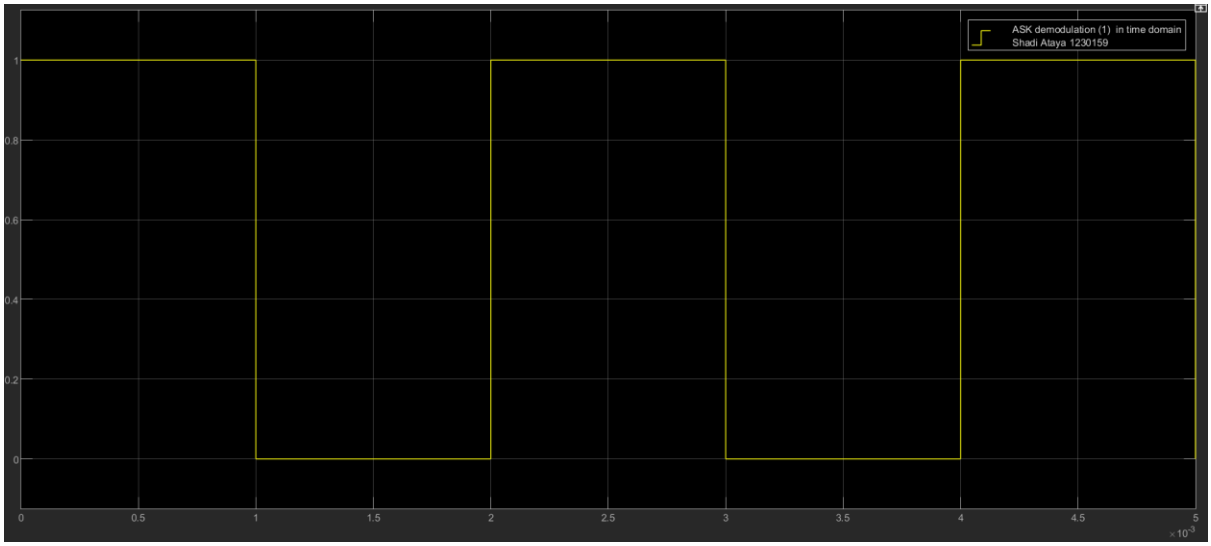


Figure 28: De-Modulated (1) signal @ Am=5, D=50% & f=0.5KHz in time domain

Frequency Domain

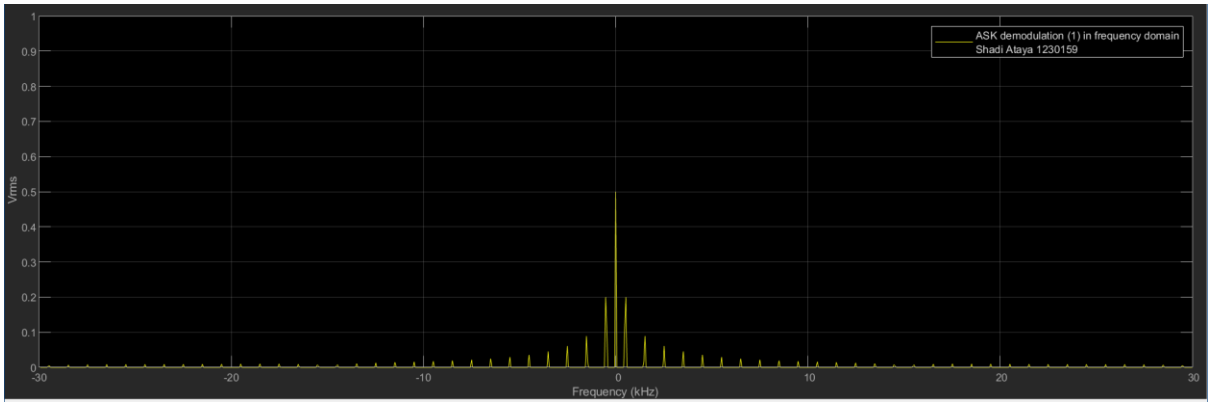


Figure 29: De-Modulated (1) signal @ Am=5, D=50% & f=0.5KHz in frequency domain

De-Modulated (1) signal @ Am=5, D=10% & f=1KHz

Time Domain

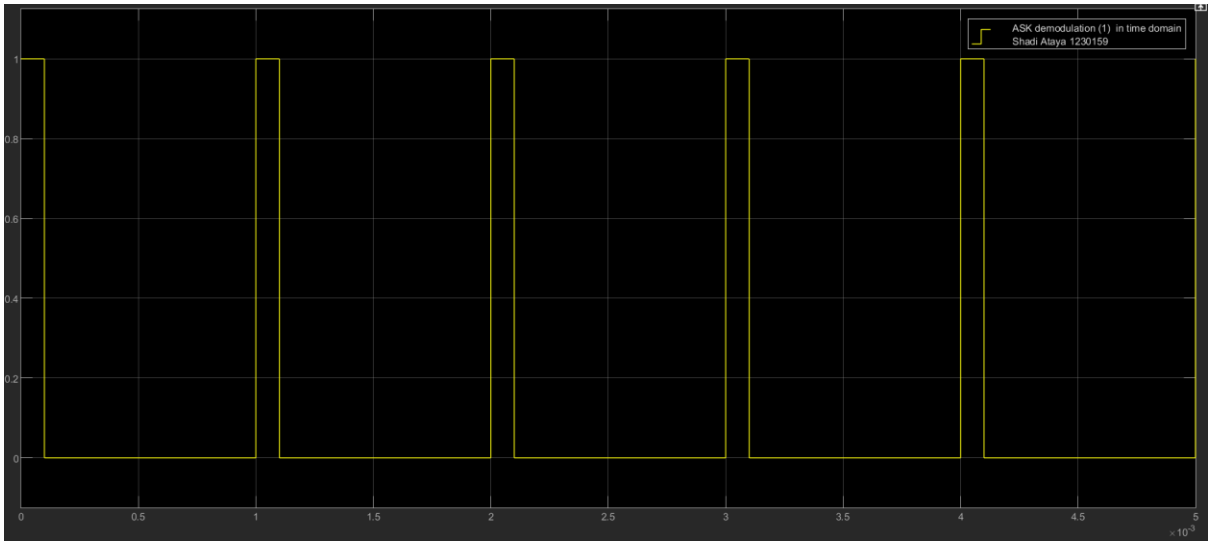


Figure 30: De-Modulated (1) signal @ Am=5, D=10% & f=1KHz in time domain

Frequency Domain

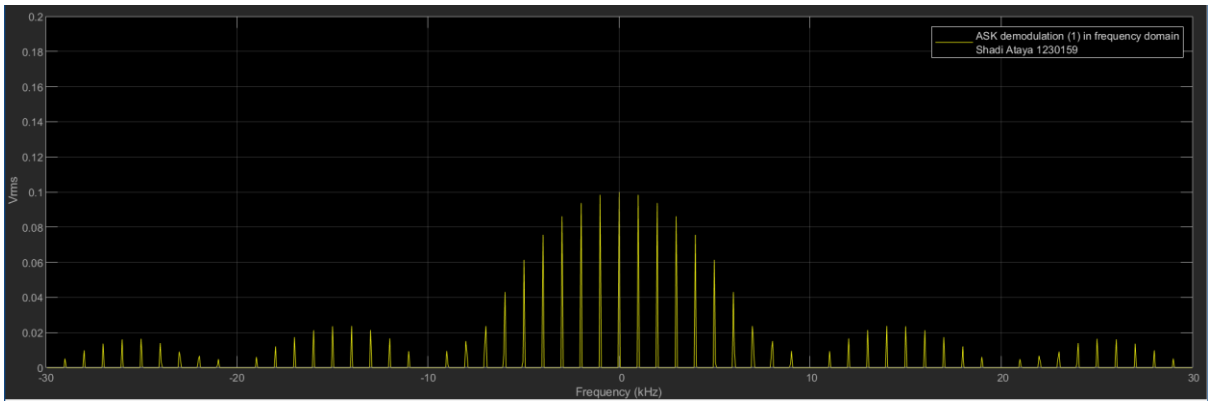


Figure 31: De-Modulated (1) signal @ Am=5, D=10% & f=1KHz in frequency domain

- In ASK Demodulation (1) for all cases, in time domain the demodulated signal has the same overall shape and timing as the original message pulse, because multiplying the ASK modulated signal by the carrier and then applying the SIGN and ABS operations effectively recovers the envelope. The only noticeable change is that its amplitude is lower than the original, since the demodulation process reduces the recovered signal level. The rising and falling edges occur at the same instants as in the message, so the ON–OFF pattern is preserved. In the frequency domain, the demodulated signal contains the same fundamental frequency and harmonic components as the original message, but all spectral lines appear at a lower amplitude than the original, reflecting the amplitude loss introduced during demodulation. Thus, the demodulated output matches the message in structure but appears simply reduced in amplitude.

ASK Demodulation 2

Block Diagram

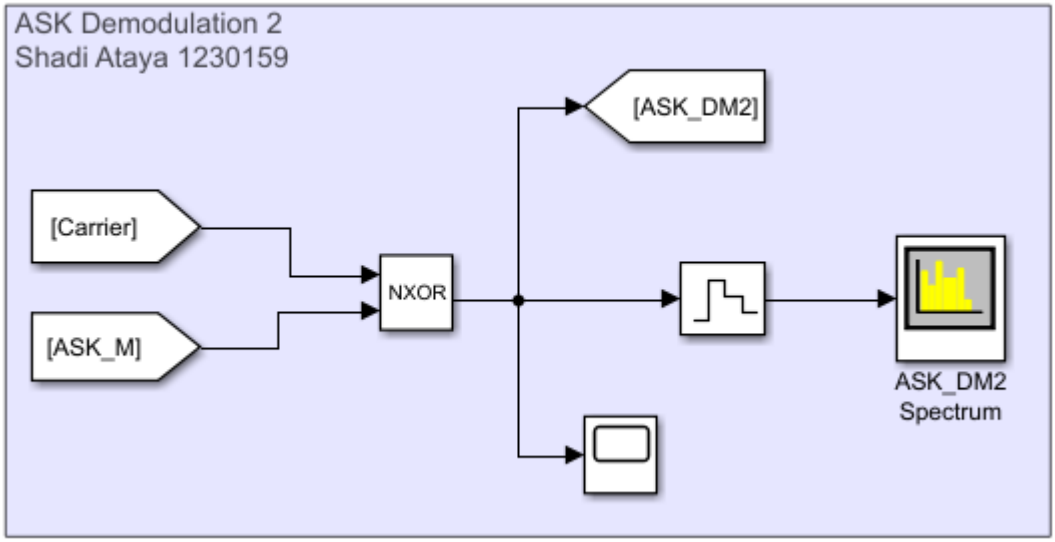


Figure 32: ASK Demodulation (2) Block Diagram'

De-Modulated (2) signal @ Am=5, D=50% & f=1KHz

Time Domain

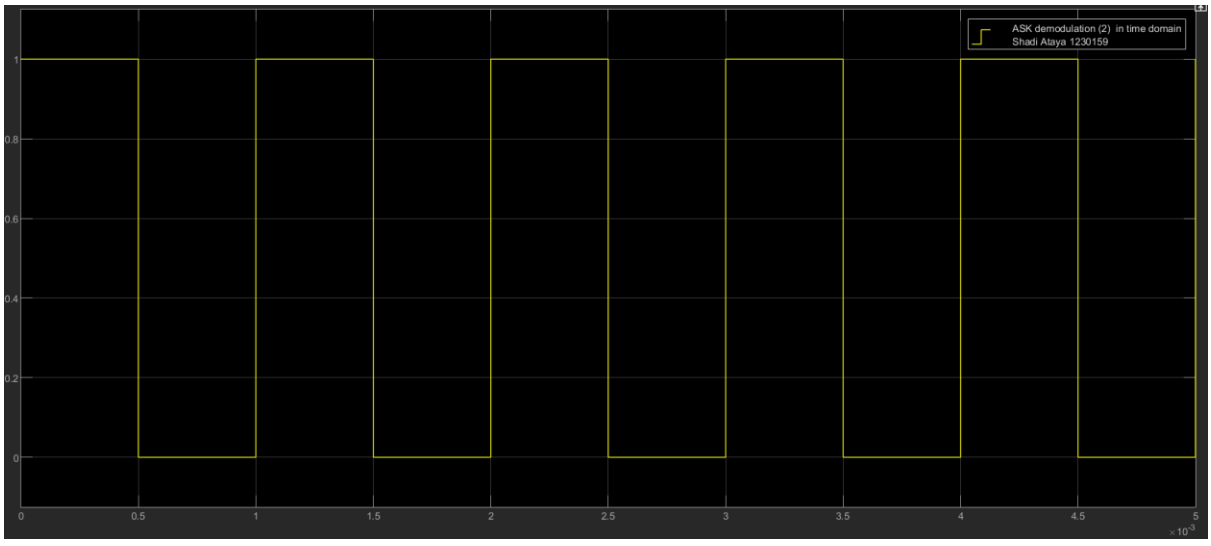


Figure 33: De-Modulated (2) signal @ Am=5, D=50% & f=1KHz in time domain

Frequency Domain

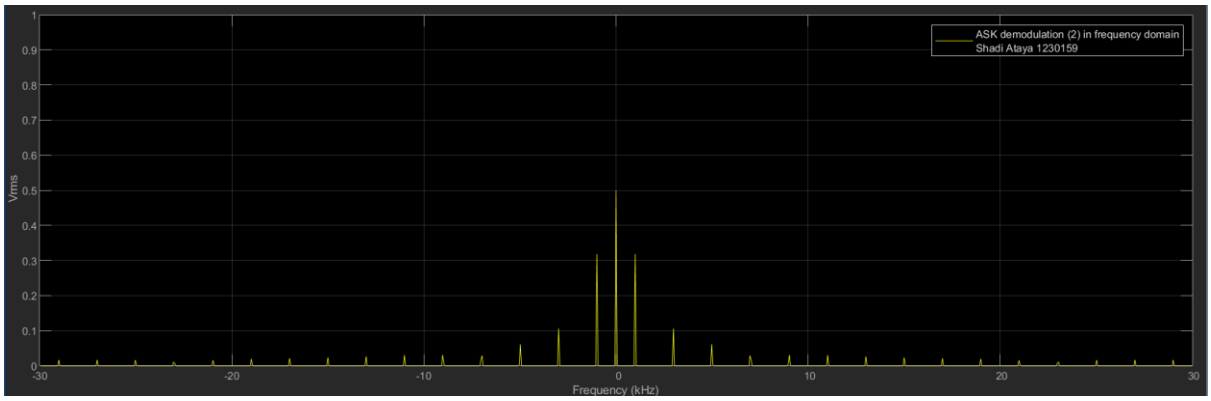


Figure 34: De-Modulated (2) signal @ Am=5, D=50% & f=1KHz in frequency domain

De-Modulated (2) signal @ Am=1.5, D=50% & f=1KHz

Time Domain

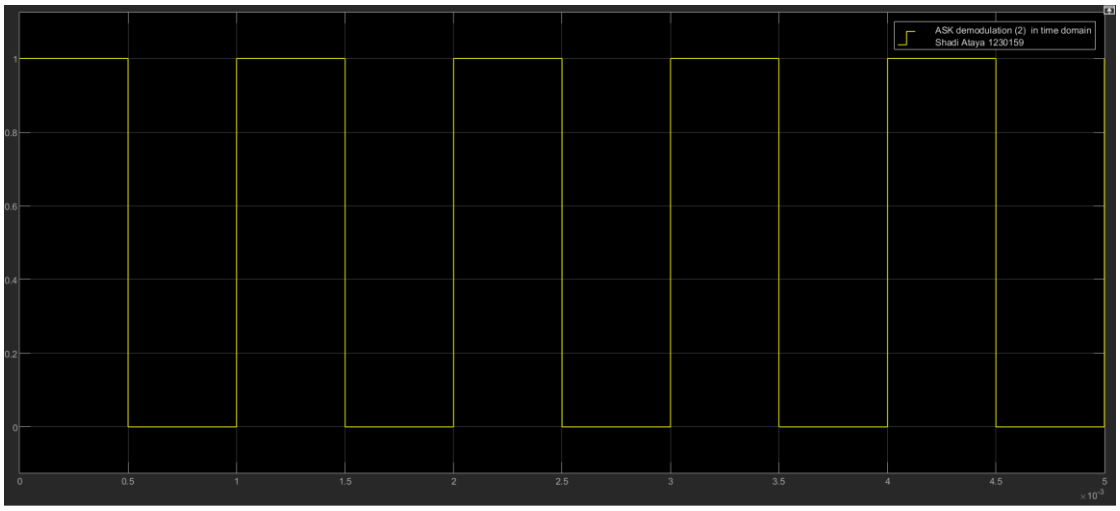


Figure 35: De-Modulated (2) signal @ Am=1.5, D=50% & f=1KHz in time domain

Frequency Domain

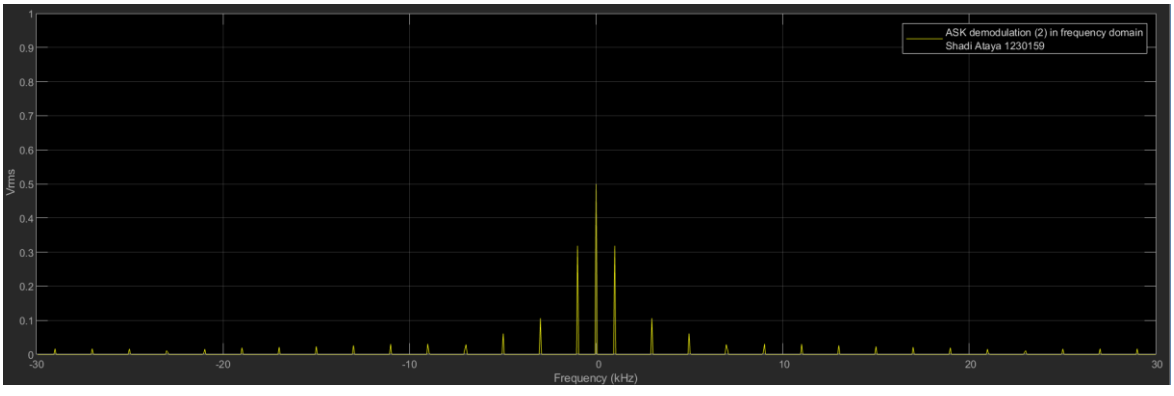


Figure 36: De-Modulated (2) signal @ Am=1.5, D=50% & f=1KHz in frequency domain

De-Modulated (2) signal @ Am=5, D=50% & f=0.5KHz

Time Domain

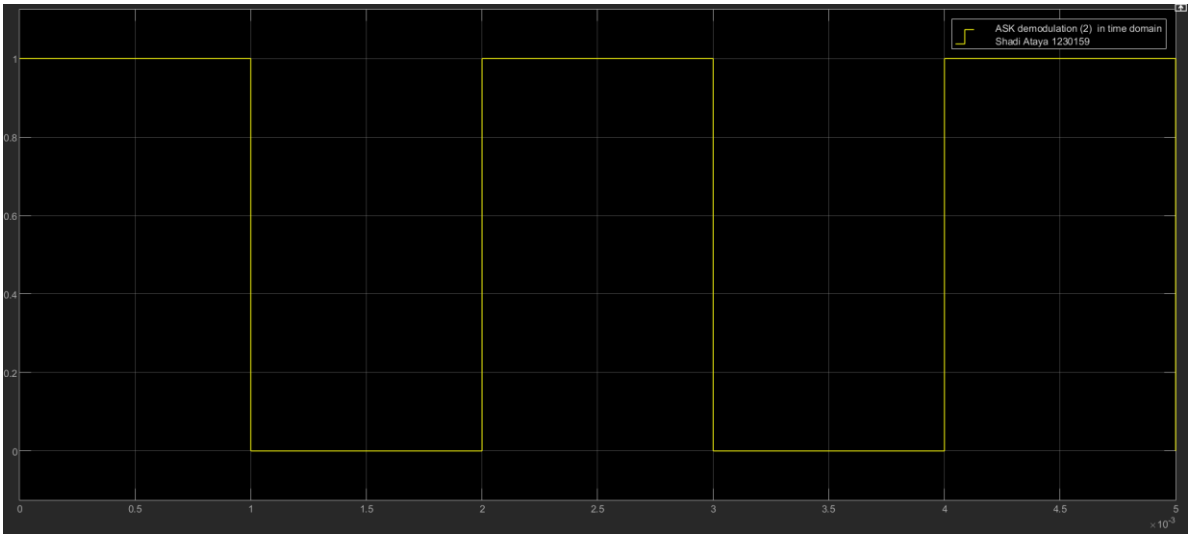


Figure 37: De-Modulated (2) signal @ Am=5, D=50% & f=0.5KHz in time domain

Frequency Domain

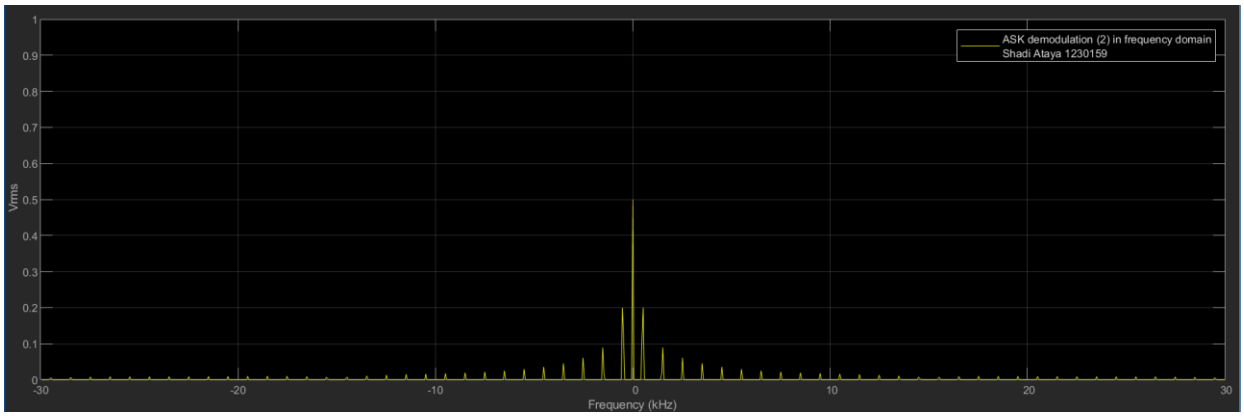


Figure 38: De-Modulated (2) signal @ Am=5, D=50% & f=0.5KHz in frequency domain

De-Modulated (2) signal @ Am=5, D=10% & f=1KHz

Time Domain

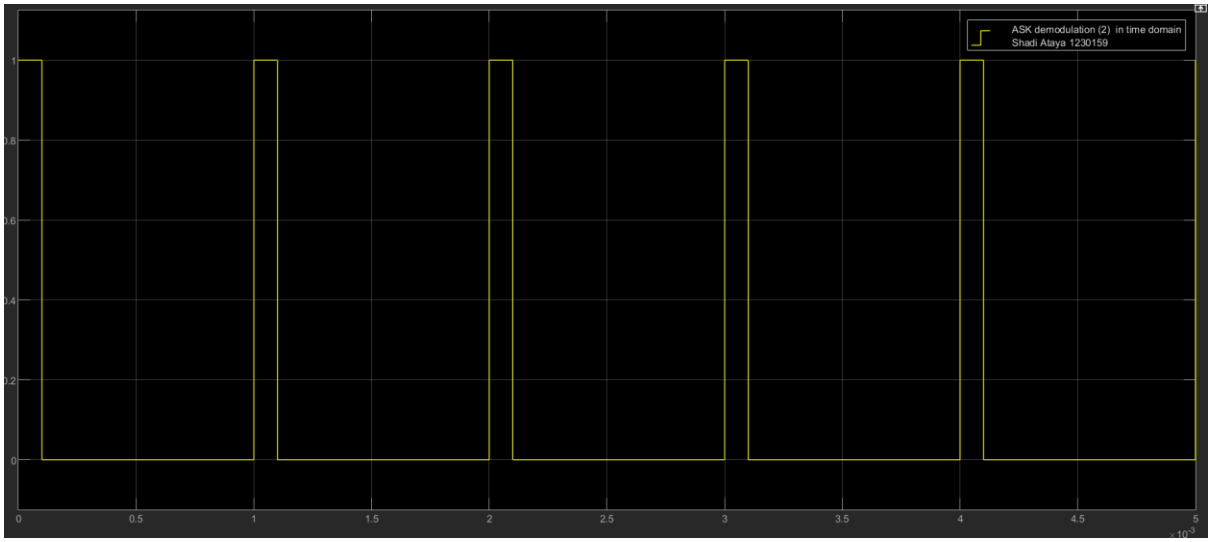


Figure 39: De-Modulated (2) signal @ Am=5, D=10% & f=1KHz in time domain

Frequency Domain

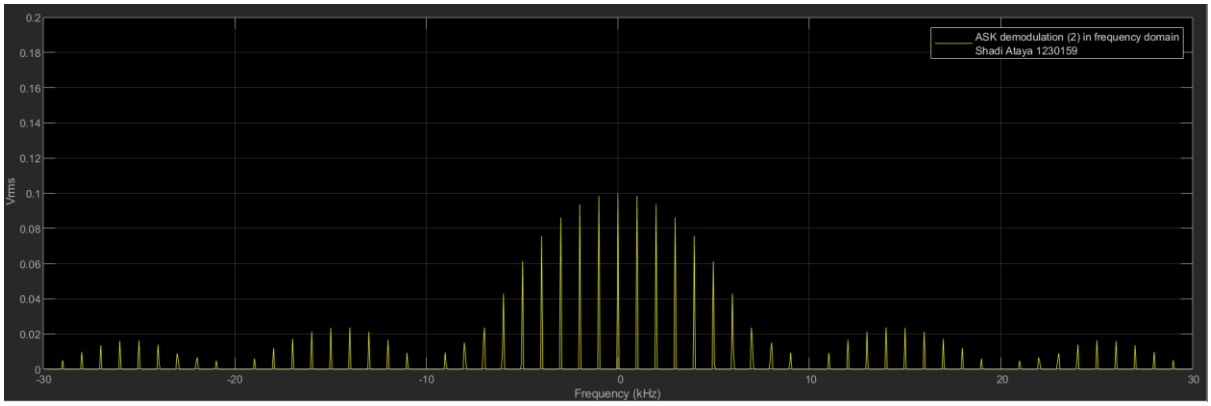


Figure 40: De-Modulated (2) signal @ Am=5, D=10% & f=1KHz in frequency domain

- In ASK Demodulation (2), after multiplying the ASK-modulated signal by the carrier, the result is applied to an NXOR operation. This effectively recovers the envelope by producing a high level whenever the multiplied signal matches the carrier's polarity, so the time-domain output almost perfectly follows the original message. for all cases the recovered pulse has the same shape, timing, and ON–OFF pattern as the message, with only a lower amplitude due to the demodulation process. In the frequency domain, the output contains the same fundamental and harmonic components as the message but with reduced magnitude. showing that the ASK Demodulation (2) method produces the same behavior as ASK Demodulation (1) method. (**Note:** Both Demodulation processes are explained step by step in the last Discussion part).

Discussion

In this pre-lab, **each part of the experiment was discussed separately under its corresponding section, and every major point was clearly highlighted using "–"** for clarity and organization. Overall, the ASK modulation was implemented by multiplying the 20 kHz carrier $\cos(2\pi \cdot 20k \cdot t)$ with the message pulse signal under different parameter settings. Reducing the message amplitude directly lowered the amplitude of the ASK-modulated waveform in the time domain, producing weaker ON intervals while the OFF intervals remained unchanged. In the frequency domain, all spectral components decreased proportionally. Reducing the duty cycle shortened the width of the ON pulses, making the modulated waveform sparser in time and increasing the harmonic content in the spectrum because of the sharper transitions. This happens because the duty cycle represents how long the signal stays "ON" in each cycle, so lowering it naturally reduces the ON duration. Lowering the message frequency increased the spacing between the ON–OFF pulses in the time domain, while in the frequency domain, decreasing fm moved the message-related spectral lines closer to the carrier, which reduces the bandwidth of the modulated signal. This is expected since decreasing fm increases the period ($T = 1/fm$), causing the ON–OFF structure to stretch in time. For demodulation, two methods were used: the first multiplied the ASK signal with the carrier and applied SIGN and ABS, where multiplying by the carrier shifts the spectrum back to baseband, the SIGN block extracts the polarity of the multiplied waveform, and the ABS block recovers the envelope by forcing all values to be positive. The second method multiplied the signal and used an NXOR block, where the NXOR compares the polarity of the multiplied signal with the carrier and outputs a high level when they match, effectively reconstructing the message envelope. Both methods recovered the original message shape in the time domain with a lower amplitude and produced the same message-related components in the frequency domain but with lower magnitudes. This reduction occurs because multiplying by the carrier and the subsequent envelope-extraction blocks introduce scaling losses. Additionally, the amplitude of the message has little effect on the demodulated output, because the demodulation process mainly needs to distinguish between ON and OFF intervals, not the exact signal height. Since both approaches produced identical results for all tested cases, the demodulation process is confirmed to be correct